MILITARY SPECIFICATION

PROPELLANT, M1
FOR USE IN CHARGE, PROPELLING, M67
FOR 105MM HOWITZER

1. SCOPE

1.1 This specification covers M1 Propellant for one type of propelling charge designated as M67 for the 105MM Howitzer, M2A2.

1.2 Classification. The propellant shall be of the following types as specified:

Type I - Cylindrical Multi-perforated Grains
Type II - Cylindrical Single-perforated Grains

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons General Specification for

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Defintions
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-652 - Propellants Solid for Cannons Requirements and Packing

FSC: 1375

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MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1171 - Acceptance and Description Sheets for Propellants and Explosives

DRAWINGS

ARMY

9288695 - Charge Propelling, M67 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation: Parts 0-199

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

(The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite "49 CFR 0-199 (latest revision)").

3. REQUIREMENTS

3.1 Materials. The propellant shall be M1, Type I, with basic lead carbonate for the zones 3 through 7 and Type II for zones 1 and 2 conforming to the requirements of Standard MIL-STD-652 and Drawing (dwg.)9288695.

3.2 Proving Ground. The propellant, when fired in the M2A2 howitzer and with the weight of projectile (33.00 pounds (lbs)) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.

3.2.1 Assessment. At 70 degrees Fahrenheit (°F) and standard weapon conditions, the propellant shall be capable

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of charge assessment to a zone 3 velocity of 765 feet per second (fps) with an associated expected pressure of 9200 to 10,500 pounds per square inch (psi), and a zone 7 velocity of 1525 fps with an associated expected pressure of 31,200 to 39,000 psi.

3.2.2 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F shall not exceed 3.8 fps at zone 3 or 3.4 fps at zone 7.

3.2.3 Permissible Individual Maximum Pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 44,800 psi at any operating temperature (-50°F to +145°F) when fired in a standard weapon.

3.3 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information

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accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of batches of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of pounds in the lot.

g. Date submitted.

h. Eight copies of propellant description sheet SMU Form 1047 containing the applicable data (see 6.7)

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation. The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of : (a) one or more batches of Type I, propellant or (b) one or more batches of Type II, propellant, produced by one manufacturer, in accordance with the same specification or same revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of M1 propellant subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Testing

4.2.2.1 Materials (see 3.1). Each lot of propellant

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shall be sampled and tested for composition, form, dimensions and chemical-physical properties in accordance with MIL-STD-652.

4.2.2.2 Proving ground (see 3.2). A ballistic test quantity of 100 pounds of Type I and 100 pounds of Type II propellant shall be randomly selected from the lot for charge establishment and uniformity firings at ambient temperature. For the first three lots of propellant from any production run, a 7 round uniformity series shall be fired at 145 degrees Fahrenheit (°F) and minus 50°F in addition to firings required to determine compliance with 3.2 at 70°F.

4.2.2.2.1 Assessment (see 3.2.1)

4.2.2.2.1.1 Zone 3 Pressure, Major defect, Code No. 01001 (see 6.4) A lot of single perforated propellant shall be referred if the average pressure at the assessed charge weight fails to meet the applicable requirement.

4.2.2.2.1.2 Zone 7 Pressure, Critical defect, Code No. 02001. A lot of multi-perforated propellant shall be rejected if the average pressure at the assessed charge weight fails to meet the applicable requirement.

4.2.2.2.2 Velocity uniformity (see 3.2.2) Major defect, Code No. 03001. If the product of the factor (see 6.5) times the velocity standard deviation of the rounds tested at any zone in accordance with 4.3.3. exceeds the applicable requirement the lot shall be rejected.

4.2.2.2.3 Individual pressure (see 3.2.3). If any individual pressure observed during uniformity firings at any operating temperature (-50°F to +145°F) after correction for difference between as fired and recommended service charge weight exceeds the requirement, the lot shall be rejected.

4.3 Test Methods and Procedures

4.3.1 Proving ground tests. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure.

4.3.2 Charge assessment

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4.3.2.1 Charge Slope Method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity. This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.3.3). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. Increment charge weights shall be determined by firing propelling charges assembled in accordance with Dwg. 9288695 (or closed bomb firings).

4.3.2.2 Constant Slope Method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite velocity versus charge weight versus pressure curve can be established from test results generated. This curve can then be used as in 4.3.2.1 above to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.3.2.3 Closed bomb. Closed bomb firings may be substituted for ballistic firings at zones one and two for charge assessment only in accordance with MIL-STD-286, Method 801.1.

4.3.3 Uniformity series. The propellant shall be conditioned to 70°F plus or minus 2°F for at least 16 hours and shall be fired within five minutes after removal from the conditioned chamber. A seven round uniformity series shall be fired at each zone specified in 3.2.2 with propelling charge assembled in accordance with dwg. 9288695. The mean velocity, velocity standard deviation, pressure standard deviation, and mean of the maximum pressure for each seven round firing shall be calculated.

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4.3.4 Retest. If for any reason the test agency or commodity center considers that test conditions have detrimentally affected the test results, they shall declare the results invalid and perform a new test on additional charges as required.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing (Level B) and marking shall be in accordance with Standard MIL-STD-652.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1)
c. Process. Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the devisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3 Submission of test data. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-F, and one copy to ATTN: SARPA-ADEP-2, Dover, New Jersey, 07801.

6.4 Inspection code number. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 per cent upper confidence limit (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller

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sample size exceeds the true standard deviation by chance alone. The factor for a normal 7-round group is 0.69.

6.6 Tolerance. A tolerance of -5°F applies to all -50°F and 145°F requirements and a tolerance of plus or minus 2°F to all 70°F requirements.

6.7 Submission of SMU Form 1047. Information contained in propellant description sheet, SMU Form 1047, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agent is adhered to (see 4.1.1)

6.8 Surveillance sample. Samples shall be forwarded to Picatinny Arsenal, Dover, N.J. ATTN: SARPA-FR-G, for the propellant surveillance test program in accordance with MIL-STD-652.

6.9 *Proving ground test summary*

INITIAL PRODUCTION

Temp (°F)	Sample Size	Vel(fps) (assessed)	Lot Std. Dev of Vel	Perm Ind Max Press	Pressure (psi) min	Pressure (psi) max
-50	7	NA	NA	44,800	NA	NA
70 (zone3)	7	765	3.8	NA	9,200	10,500
70 (zone 7)	7	1525	3.4	44,800	31,200	39,000
145	7	NA	NA	44,800	NA	NA

SUBSEQUENT PRODUCTION

70 (zone 3)	7	765	3.8	NA	9,200	10,500
70 (zone 7)	7	1525	3.4	44,800	31,200	39,000

Custodian:
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Preparing Activity:
ARMY-PA

Project Number: 1375-A-062

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